D02 · Inventory Strategy & Optimization

05 · Inventory KPI dashboard & reporting framework

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1 WHAT IT IS

An inventory KPI dashboard and reporting framework measures inventory by composition and quality, not just total value or days — decomposing it into cycle, safety, in-transit, pre-build, and excess/obsolete, and pairing turnover and service with health metrics — so leadership can distinguish productive inventory from waste and act on the difference, instead of steering by a single aggregate number that hides what it is made of.

2 WHY IT MATTERS

Gaur, Fisher & Raman 2005 **(Gaur, Fisher & Raman, 2005 — peer-reviewed)**, analyzing inventory turnover across retailers, showed that turnover must be interpreted against the factors that drive it — margin, capital intensity, and sales surprise — because the raw number alone is not comparable or actionable. A dashboard that reports only total inventory value or aggregate days inherits this flaw at the operating level: it cannot tell whether the inventory is healthy cycle and safety stock doing its job or dead excess sitting idle. Two businesses with identical total days can have opposite inventory health. The cost is misdirected action — **a leadership team steering on the aggregate number cuts healthy buffer and ignores the excess**, because the metric never told them which was which.

3 HOW IT IS DONE

1. **Report composition, not just the total.** Break inventory into cycle, safety, in-transit, pre-build, and excess/obsolete on the dashboard. The total tells you how much; the composition tells you whether it is healthy — and only the composition supports a decision.
2. **Pair quantity metrics with quality metrics.** Show turnover and days-of-supply alongside health indicators — excess/obsolete share, aging, service achieved. Quantity without quality is the metric that lets a leadership team congratulate itself on low days while half the stock is dead.
3. **Interpret turnover against its drivers.** Turnover is only meaningful relative to margin, lead time, and demand variability. Report it with context — by segment and against a relevant benchmark — so it informs action rather than inviting a naive "turn faster" mandate that ignores why the turn is what it is.
4. **Segment the dashboard by ABC-XYZ and location.** Aggregate inventory health is the average of healthy and unhealthy pockets. Reporting by segment and location surfaces where the excess actually sits, which the total can never show. Granularity is what makes the dashboard diagnostic.
5. **Connect inventory KPIs to service and working capital.** Show inventory next to its purpose (service) and its cost (working capital, days of inventory outstanding). An inventory metric divorced from service and cash is a number with no business meaning — the trade-off is the message.
6. **Make the dashboard drive a standing review.** Tie the metrics to a recurring inventory review with owned actions — disposition, policy change, driver projects. A dashboard no one acts on is wallpaper; the reporting framework earns its keep only when it routes attention to the pockets that need it.

> **WORKED EXAMPLE — ILLUSTRATIVE**
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> A company reported inventory as a single value and a days-of-supply number — **baseline: 62 days reported as "on target," but no visibility that 22% was excess/obsolete while A/X service was being starved to hold the average**. It rebuilt the dashboard around composition, segment, and health, paired with service and working capital. Results: **the hidden excess became visible and was disposed, healthy buffer was protected, days fell from 62 to 51 with service up** — the same data, finally reported in a way that distinguished productive inventory from waste.

IN PRACTICE

A composition-first dashboard

Lead the dashboard with the inventory breakdown by component, then segment and location, then trend. The composition view is the one that drives action; total value and days belong as context, not as the headline. Designing the dashboard around composition is what makes it a decision tool rather than a report.

Pair every quantity metric with a quality and a purpose metric

Never show days or turnover without excess share and service alongside. The discipline of always pairing quantity, quality, and purpose prevents the most common reporting failure — celebrating a low number that was achieved by starving service or hiding excess.

Segment and location granularity

Report inventory health by ABC-XYZ segment and by location, with drill-down. The aggregate is an average that hides the pockets; the granularity is where the excess and the starved buffer actually live, and where action is taken. A dashboard that only aggregates cannot direct a single decision.

Tie the dashboard to a standing review

Connect the metrics to a recurring inventory review with owned actions and a feedback loop to policy and forecasting. A dashboard that informs no decision is decoration; the value of the reporting framework is realized only when it routes leadership attention to the pockets that need it, cycle after cycle.

WHAT YOU WOULD USE

INVENTORY KPI / REPORTING & ANALYTICS

SAP IBP / o9 — embedded inventory analytics — Native inventory decomposition, health, and turnover analytics on the planning data, with configurable dashboards and drill-down by segment and location. **(SAP, o9 — vendor)** The choice when inventory reporting should sit on the same model that sets policy and runs the plan.

Kinaxis / Blue Yonder / RELEX analytics — Strong embedded inventory dashboards covering composition, aging, and service trade-offs. **(Kinaxis, Blue Yonder, RELEX — vendor)** A solid fit where the reporting rides on the planning and replenishment engine already in place.

BI on an inventory data mart (Power BI, Tableau) — BI platforms built on an inventory data mart, giving full control over composition definitions, health metrics, and benchmarks. **(BI vendors — vendor)** The vertical option when the dashboard must be tailored beyond what the APS reports natively.

For operations that don't need a platform — A composition-first inventory report in a spreadsheet — components, segment, location, health, paired with service and working capital — reviewed on a fixed cadence. The decisive choice is reporting composition over a single total; that reframing, not the visualization tool, is what turns the dashboard into a decision.

4 SO WHAT

Operations whose dashboard reports composition and health steer toward the excess and protect the healthy buffer; operations that report a single aggregate number cut blindly, trimming the safety stock that holds service while the dead excess sits untouched. The aggregate inventory number is the most comfortable and least actionable metric on the scorecard — it grades the quantity and says nothing about the quality. A dashboard that cannot separate productive inventory from waste is not measuring inventory; it is measuring a pile.

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